AKHIL JAINI

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PROFESSIONAL SUMMARY

I am a radio astronomer and PhD candidate at Swinburne University specialising in fast radio bursts (FRBs), interferometry, and real-time transient detection. My research focuses on developing pipelines for transient detection and improving astrometric precision with ASKAP to sub-arcsecond levels, enabling robust host-galaxy associations. I have extensive experience in computational methods, high-performance data processing, and astronomical instrumentation, supported by multiple first-author publications, invited seminars, and contributions to major collaborations. I thrive in collaborative, data-intensive research environments and am particularly interested in extending my research to VLBI-scale localisation and leveraging large FRB samples to constrain their origins and environments.

ACADEMIC DETAILS, SKILLS, AND INTERESTS

CERTIFICATE	SPECIALIZATION	UNIVERSITY/BOARD	RESULT	YEAR
PhD	Astrophysics	Swinburne University of Technology, Melbourne, Australia	--	2023 - Ongoing
MTech	Astronomical Instrumentation	Indian Institute of Astrophysics, Bangalore, India and University of Calcutta, Kolkata, India	CGPA: 9.2/10	2020 - 2022
BE (Hons)	Electrical & Electronics	Birla Institute of Technology and Science - Pilani, India	CGPA: 7/10	2014 - 2018

Research Interests: Radio Astronomy, Computational Astrophysics, Astronomical Instrumentation.

Programming Languages: Python, C, C++, CUDA, HTML, JavaScript.

Tools: Linux, OzSTAR/NT (HPC), Git/GitHub, Docker, LaTeX

REFEREED PUBLICATIONS

- **Jaini, A.,** Deller, A. T., Wang, Y, … & Lenc, E. (Submitted). Enhanced Astrometry of the Rapid ASKAP Continuum Survey: Mid and High Frequency Epochs. *arXiv pre-print.* arXiv
- **Jaini, A.,** Deller, A. T., Wang, Y, … & Lenc, E. (2025). Enhanced Astrometry of the Rapid ASKAP Continuum Survey for Precise Localisation of Fast Radio Bursts. *Publications of the Astronomical Society of Australia,* 42, e060. doi:10.1017/pasa.2025.28 | arXiv
- **Jaini, A.,** Deshpande, A., & Bitragunta, S. (2021). A minimal space interferometer configuration for imaging at low radio frequencies. *Publications of the Astronomical Society of Australia, 38*, E040. doi:10.1017/pasa.2021.34 | arXiv

Full list of publications can be found at: ui.adsabs.harvard.edu/public-libraries/6AWkqcbzTQWau_usyl5V8Q

COLLOQUIA/INVITED TALKS

- **Colloquium** at Curtin Institute of Radio Astronomy (CIRA), Curtin University, Perth, Australia on *Detecting and Localising Fast Radio Bursts (and other Transients) with ASKAP* | **March 2026**
- **Invited Seminar** at the Indian Institute of Astrophysics (IIA), Bengaluru, India on *Detecting and Localising Fast Radio Bursts (and other Transients) with ASKAP* | **December 2025**
- **Colloquium** at the Department of Space, Planetary & Astronomical Sciences & Engineering (SPASE), Indian Institute of Technology – Kanpur (IITK), Kanpur, India on *Advances in FRB Detection and Astrometry using ASKAP* | **December 2025**

RESEARCH EXPERIENCE

PhD Scholar – Centre for Astrophysics and Supercomputing, Swinburne University, Melbourne, Australia. **June 2023-Ongoing**

Guide: *Prof. Adam Deller, Associate Professor, Centre for Astrophysics and Supercomputing (CAS), Swinburne University of Technology.*
Co-Guide: *Dr. Yuanming Wang, Postdoc Fellow, Centre for Astrophysics and Supercomputing (CAS), Swinburne University of Technology.*
Co-Guide: *Prof. Ryan Shannon, Associate Professor, Centre for Astrophysics and Supercomputing (CAS), Swinburne University of Technology.*

Project: Probing the Cosmic Web with Fast Radio Bursts

- Developing a real-time Fast Radio Burst (FRB) detection pipeline for the Commensal Real-time ASKAP Fast Transient (CRAFT) Coherent (CRACO) upgrade to the Australian Square Kilometre Array Pathfinder (ASKAP) telescope.
- Improving the localising precision of current and future ASKAP FRBs by cross-referencing the data with other ASKAP data products, and by utilizing data from ASKAP's parallel continuum imaging system.
- Characterising the Rotating Radio Transients (RRATs) discovered by ASKAP CRACO and following up their periodicity and polarisation properties with the Murriyang/Parkes Telescope.

Junior Research Fellow – Indian Institute of Astrophysics (IIA), Bangalore, India. **August 2022-April 2023**

Guide: *Prof. Sivarani Thirupathi, Professor, Indian Institute of Astrophysics (IIA).*
Co-Guide: *Prof. S. P. Rajaguru, Professor, Indian Institute of Astrophysics (IIA).*

Project: Reimagining Existing and Future Spectrographs with a focus on Exoplanet Transits

- Understood the characteristics and corrected for the sources of systematic errors of existing 2m class telescopes during high- and low-resolution spectrophotometric observations of exoplanet transits and extrapolated them for future spectrographs.

MTech Student – Indian Institute of Astrophysics (IIA), Bangalore, India. **December 2021-September 2022**

Guide: *S. Sriram, Engineer-F and Head, Optics Division, Indian Institute of Astrophysics (IIA).*
Co-Guide: *Prof. Annapurni Subramaniam, Director, Indian Institute of Astrophysics (IIA).*

Project: Design and Development of Digital Micromirror Device (DMD) based Multi-Object Spectrograph (MOS) for INSIST

- Designed and simulated the ultraviolet spectrograph with the required optical components, developed a Python GUI package for calibration and control of the spectrograph by the end user, and tested the spectrograph under laboratory conditions.

Visiting Research Student - Raman Research Institute (RRI), Bangalore, India. **January 2018-March 2020**

Guide: *Prof. Avinash Deshpande, Professor, Astronomy and Astrophysics Division, Raman Research Institute (RRI).*
Co-Guide: *Prof. Sainath Bitragunta, Assistant Professor, Department of Electrical and Electronics Engineering, BITS Pilani.*

Project: Simulation of a Minimal Space-Interferometer-Configuration for Low Frequency Radio Observations

- Developed a Python simulation package for a minimal system of radio telescopes aboard low earth orbit satellites and calculated various parameters to determine the advantages such a system would have in terms of all-sky coverage at low radio frequencies.

Summer Research Student - National Centre for Radio Astrophysics (NCRA-TIFR), Pune, India. **May 2017-July 2017**

Guide: *Prof. Yashwant Gupta, Director, National Centre for Radio Astrophysics (NCRA-TIFR).*

Project: Designing a Real-Time Radio Transient Detection Pipeline for the upgraded Giant Metrewave Radio Telescope (uGMRT)

- Developed a pipeline for detection of radio transient phenomena by porting previously written code into CUDA C, writing new and relevant parallelized codes where necessary, and tested the pipeline by simulating the uGMRT data flow.

SCIENTIFIC CONFERENCES

- **Jaini, A. (Speaker)** (2026, July). *Origins of FRB Scattering: Characterising FRB Scattering Timescales via Host Galaxy Sub-structure* [Contributed Talk]. Fast Radio Bursts 2026 (FRB2026), National Astronomical Observatories, Chinese Academy of Sciences, Guiyang, China. https://frb2026.casconf.cn/page/1958752265951121409
- **Jaini, A. (Speaker)** (2026, February). *Precision by Design: Astrometric Pathways for Post-2030 Widefield Arrays* [Contributed Talk]. Australia Telescope National Facility (ATNF) Futures 2030 Workshop (online), Commonwealth Science and Industrial Research Organisation (CSIRO), Sydney, Australia. https://www.atnf.csiro.au/event/atnf-futures-2030-workshop/
- **Jaini, A. (Speaker)**, Deller A. T., Wang, Y. (2025, July). *Enhanced astrometry of the Rapid ASKAP Continuum Survey for precise localisation of FRBs* [Contributed Talk]. Astronomical Society of Australia – Annual Science Meeting (ASA ASM 2025), University of Adelaide, Adelaide, Australia. https://indico.global/event/12673/contributions/128435

- **Jaini, A. (Speaker)**, Deller, A. T., Wang, Y. (2024, November). *Precise pinpointing of FRBs using ASKAP* [Flash Talk]. Fast Radio Bursts 2024 (FRB2024), National Astronomical Research Institute of Thailand (NARIT), Khao Lak, Thailand. https://indico.narit.or.th/event/204/page/833-programme
- **Jaini, A. (Speaker)**, Deller, A. T., Wang, Y. (2024, September). *Precise pinpointing of FRBs using ASKAP* [Contributed Talk]. Mount Stromlo Student Seminar 2024 (MSSS 2024), Australian National University (ANU), Canberra, Australia. https://www.mso.anu.edu.au/msss24/#Program
- **Jaini, A. (Speaker)**, Deller, A. T., Wang, Y. (2024, June). *Precise pinpointing of FRBs using ASKAP* [Sparkler & Recorded Talk]. Astronomical Society of Australia – Annual Science Meeting (ASA ASM 2024, online), International Centre for Radio Astronomy Research, Perth, Australia. https://www.icrar.org/conferences/asa2024/abstracts/
- **Jaini, A. (Speaker)**, Sivarani, T. (2023, March). *Slitless Spectrophotometry of Exoplanet Host Stars* [Contributed Talk]. 2023 Astronomical Society of India Conference (ASI 2023), Indian Institute of Technology (IIT-Indore), Indore, India. https://www.astron-soc.in/asi2023/sites/default/files/bp_file_uploads/ASI2023_abstract_book1.pdf
- **Jaini, A. (Speaker)**, Sriram, S., Subramaniam, A. (2022, November). *Assessing the Performance of a Digital Micromirror Device based Multi-Object Spectrograph for the Indian Spectroscopic and Imaging Space Telescope* [Contributed Talk]. Young Astronomers' Meet (YAM 2023), Aryabhatta Research Institute of Observational Sciences (ARIES), Nainital, India. https://www.aries.res.in/yam2022/YAM_handbook.pdf
- Hoodati, A. (Speaker), **Jaini, A.**, Della, V., Sriram, S. (2022, September). *Assessing the Performance of a Digital Micromirror Device based Multi-Object Spectrograph for the Indian Spectroscopic and Imaging Space Telescope* [Contributed Talk]. Annual Conference on Modern Engineering Trends in Astronomy (META 2020), Indian Institute of Astrophysics (IIA), Bangalore, India. https://events.iiap.res.in/event/1/contributions/29/contribution.pdf
- **Jaini, A. (Speaker)**, Deshpande, A., Bitragunta, S. (2020, February). *Simulating a Minimal Space Interferometer Configuration for Low Frequency Radio Imaging* [Contributed Talk]. 2020 Union Radio-Scientifique Internationale – Regional Conference on Radio Science (URSI-RCRS 2020), Indian Institute of Technology (IIT-BHU), Varanasi, India. https://conferences.iitbhu.ac.in/URSI-RCRS2020/Final%20Booklet.pdf
- **Jaini, A. (Speaker)**, Chandrahasa, J. S. G., Bitragunta, S. (2018, September). *Best Reference Antenna Selection-based Radio Frequency Interference Mitigation Scheme for Radio Astronomy* [Contributed Talk]. Annual Conference on Modern Engineering Trends in Astronomy (META 2018), National Centre for Radio Astrophysics (NCRA-TIFR), Pune, India. Conference Proceedings

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- **Jaini, A. (Presenter)** (2026, July). Deller, A. T., Wang, Y. (2026, March). *Precisely pinpointing FRBs: Sub-arcsecond astrometry with ASKAP* [Conference Poster]. Fast Radio Bursts 2026 (FRB2026), National Astronomical Observatories, Chinese Academy of Sciences, Guiyang, China. https://frb2026.casconf.cn/page/1958752265951121409
- **Jaini, A. (Presenter)**, Deller, A. T., Wang, Y. (2026, March). *Precisely pinpointing FRBs: Sub-arcsecond astrometry with ASKAP* [Conference Poster, **won Best Poster Award**]. Australia-China Consortium of Astrophysical Research (ACAMAR 11) Workshop, University of Western Australia (UWA), Geraldton, Australia. https://acamar.org.au/acamar-11-posters-and-videos/
- Manjunath, B. (Presenter), Sivarani, T., Unni, A., **Jaini, A.**, Shivangi, M, Parvathy, M. (2023, August). *Exploring the feasibility of transmission spectroscopy with Hanle Echelle Spectrograph* [Conference Poster]. Strange New Worlds: The Exploration of ExoPlanets, Indian Institute of Science Education and Research (IISER-Pune), Pune, India.
- Sriram, S. (Presenter), Vineeth, V., Subramaniam, A., Maheshwar, G., Chand T., Unni V. C., **Jaini, A**. (2022, March). *Digital Micromirror Device based Multi Object Spectrograph for Indian Spectroscopic and Imaging Space Telescope* [Conference Poster]. 2022 Astronomical Society of India Conference (ASI 2022), Indian Institute of Technology (IIT-Roorkee), Roorkee, India. https://astron-soc.in/asi2022/abstract_details/ASI2022_745

RESEARCH SCHOOLS/WORKSHOPS

- **Harley Wood School of Astronomy 2024** at the University of Sydney, Sydney, Australia | **June 2024**
- **CSIRO Space and Astronomy Radio School 2023** at the Australian Telescope Compact Array (ATCA) Radio Observatory, Narrabri, Australia | **September 2023**
- **Training on Log-Periodic Dipole Antenna Design and Band-Stop Filter Design** at the Gauribidanur Radio Observatory, Indian Institute of Astrophysics (IIA), Gauribidanur, India | **November 2021**
- **Training on High-Resolution Spectrograph Design** at the IIA Optics Lab, Indian Institute of Astrophysics (IIA), Bangalore, India | **October 2021**
- **Hands-On with 1.3m J. C. Bhattacharya Telescope (JCBT), 2.3m Vainu Bappu Telescope (VBT) and CCD Gain Calibration** at the Vainu Bappu Observatory, Indian Institute of Astrophysics (IIA), Kavalur, India | **September 2021**
- **Hands-On Experience with the Sky Watch Array Network (SWAN) – The Indian effort towards a nationwide VLBI network** at the Gauribidanur Radio Observatory, Raman Research Institute (RRI), Gauribidanur, India | **July 2016**

OBSERVING PROPOSALS

- Wang, Y. (Principal Investigator), **Jaini, A.,** … (Parkes-2025OCTS / P1345). Awarded 60+ hours with the Murriyang/Parkes Telescope for observing proposal, *Follow up of rotating radio transients discovered with ASKAP CRACO.* | October 2025-September 2026
- **Jaini, A. (Principal Investigator)**, Sivarani, T. (VBT-2023-C2-P10). Awarded 3 full nights with the 2.3m Vainu Bappu Telescope (VBT) for observing proposal, *High-resolution Transit Spectroscopy for Exoplanet Transits*. | May-August 2023
- **Jaini, A. (Principal Investigator)**, Sivarani, T. (HCT-2023-C2-P54). Awarded 3 full nights with the 2m Himalayan Chandra Telescope (HCT) for observing proposal, *Transit Spectroscopy in Slit-less Mode for Exoplanet Host Stars*. | January-August 2023

OTHER RELEVANT PROJECTS

- **A Study of Physical Properties of Ionized Regions in Galactic H-II Regions** guided by *Prof. Kaushar Vaidya, Assistant Professor, Department of Physics, BITS Pilani* | **August 2017-December 2017**
- **Design and Construction of a 3.7m Parabolic Dish and Horn Antenna for Radio Astronomy** guided by *Prof. Praveen Kumar A. V., Assistant Professor, Department of Electrical and Electronics, BITS Pilani* | **January 2017-December 2017**
- **Design of a Radio Frequency Interference Mitigation Scheme for Radio Astronomy Antennas** guided by *Prof. Sainath Bitragunta, Assistant Professor, Department of Electrical and Electronics, BITS Pilani* | **January 2017-May 2017**

ACHIEVEMENTS AND AWARDS

AWARD/ACHIEVEMENT	AWARDED BY	YEAR
ASA Student Travel Assistance Scheme	Astronomical Society of Australia	2026
ACAMAR 11 Best Poster Award (Australian)	ACAMAR 11 SOC/City of Greater Geraldton	2026
ACAMAR 11 Travel Grant	ACAMAR 11 SOC/City of Greater Geraldton	2026
CAS Student Travel Fund	Centre for Astrophysics and Supercomputing	2024
IIA Travel Grants (Multiple)	Indian Institute of Astrophysics	2022-2023
I. J. Nagrath Student Project Fund	Birla Institute of Technology and Science - Pilani	2016
Nehru Memorial Scholarship for Students with Exceptional Merit	National Aluminium Company Ltd, Government of India	2014-2017

POSITIONS OF RESPONSIBILITY

- **Educator,** AstroTours: Immersive 3D astronomy experiences for the public (October 2023-Current)
- **Mentor,** STEMPals: A platform to connect enthusiastic school students to STEM professionals (January 2024-Current)
- **Outreach Volunteer,** OzGrav and the Centre for Astrophysics and Supercomputing (June 2023-Current)

- **Host,** Cookies ‘n’ Code: A weekly session for code and career talks while enjoying homemade cookies (March 2025-February 2026)
- **Mentor,** ASTRAL Institute: A platform to develop a cohort of inspirational student STEM leaders (January 2025-December 2025)
- **Host,** CAS Journal Club: A weekly session to discuss new and relevant research papers in astronomy (April 2024-March 2025)
- **Coordinator,** CAS Social Activities: Organised monthly social activities for PhD students (February 2024-January 2025)
- **Lab Assistant,** Swinburne University: Guided students through the Energy and Motion lab course (January 2024-May 2024)
- **Writer and Editor,** CosmicVarta: A platform for promoting Indian astronomy research (April 2022-December 2024)
- **Mentor (and Ex-Member),** Sky Watch Array Network (SWAN), Raman Research Institute (August 2021-September 2023)
- **Freelance Writer, Editor and Graphic Designer,** Fiverr (June 2020-May 2023)
- **Writer and Editor,** DOOT: Half-yearly magazine of the Indian Institute of Astrophysics (October 2022-April 2023)
- **Member,** Science Communication, Public Outreach & Education (SCOPE), Indian Institute of Astrophysics (January 2022-April 2023)
- **Physics and Mathematics Tutor**, HashLearn (currently BYJU’S; May 2016-June 2020)
- **Co-Founder and Coordinator**, The Radio Astronomy Club (TRAC), BITS Pilani (April 2016-December 2017)
- **Science Event Coordinator**, Astro Club, BITS Pilani (April 2016-July 2017)
- **Core Member,** Photography Club, BITS Pilani (August 2014-July 2017)